



**Escola Superior
de Tecnologia
e Gestão**

Politécnico de Coimbra

Inteligência Artificial

Descriptive Statistics

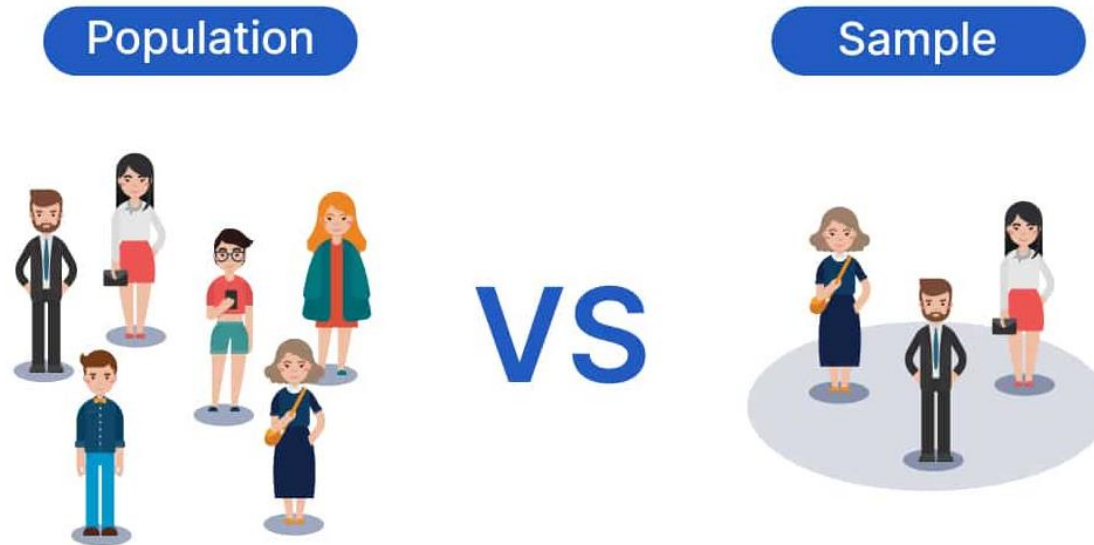
Table of Contents

- **Statistics**
- **Measures of Central Tendency**
 - Mean
 - Median
 - Mode
- **Measures of Dispersion**
 - Standard Deviation
 - Quartile
 - Interquartile Range (IQR)
- **Measures of Distribution**
 - Skewness
 - Kurtosis
- **SPSS**

- **Statistics**
- the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data.
 - **descriptive statistic**: summary statistic that quantitatively describes or summarizes features of a collection of information. Descriptive statistics aims to summarize a sample, rather than use the data to learn about the population that the sample of data is thought to represent.
 - **inferential statistics (or inductive statistics)**: infers properties of a population, for example by testing hypotheses and deriving estimates. It is assumed that the observed data set is sampled from a larger population. Descriptive statistics is solely concerned with properties of the observed data, and it does not rest on the assumption that the data come from a larger population.

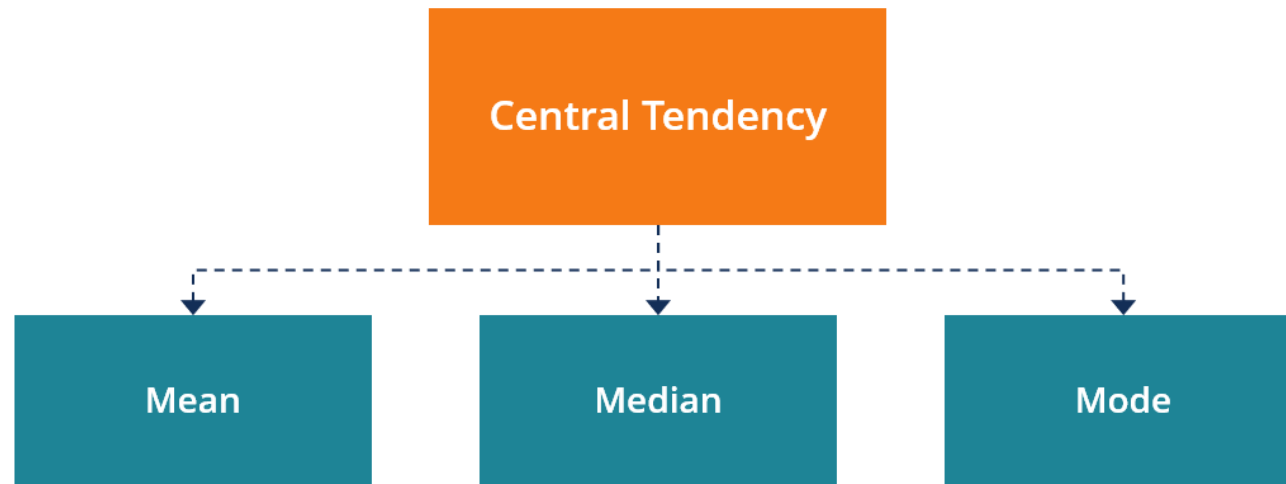
Statistics

- **Population and sample**
- a **sample** is a set of individuals or objects collected or selected from a statistical **population** by a defined procedure. The elements of a sample are known as sample points, sampling units or observations.



Descriptive Statistics

- **Measures of Central Tendency**
 - **Mean**
 - **Median**
 - **Mode**



Descriptive Statistics

- **Measures of Central Tendency**
- **Mean** (or simply mean): of a list of numbers, is the sum of all of the numbers divided by the number of numbers.

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N}$$

DATA: { 16, 17, 10, 13, 20, 18, 13, 14, 18 }

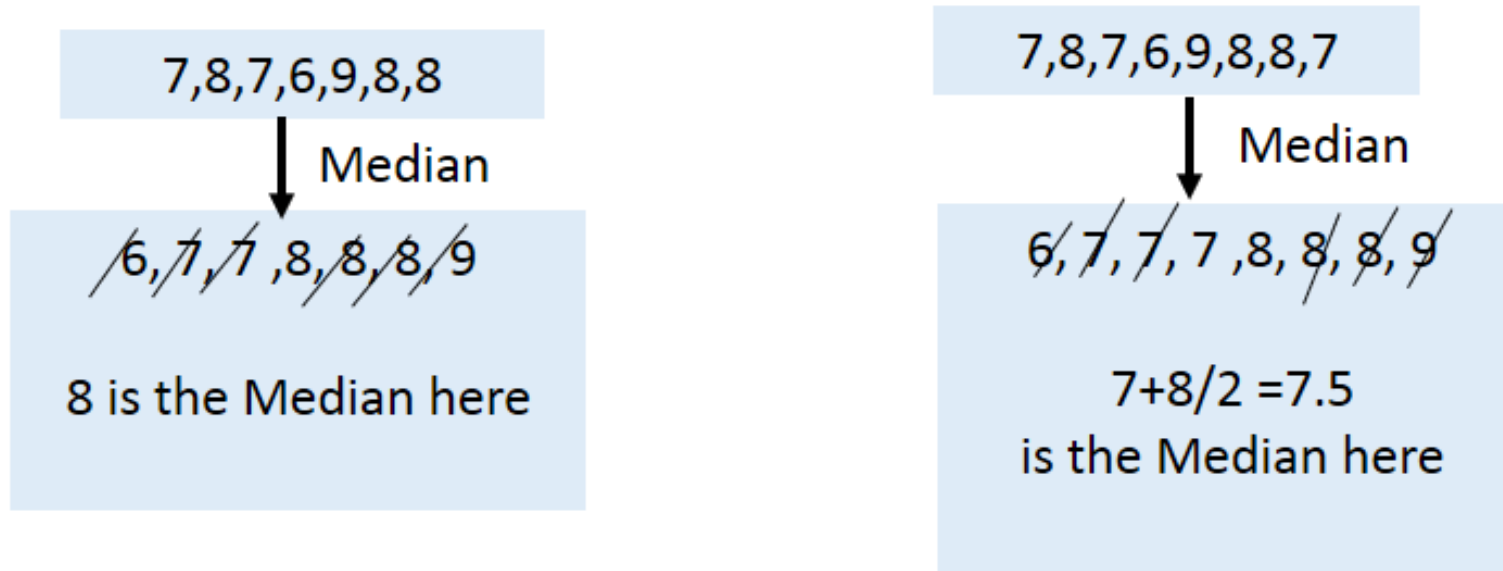
$$\bar{x} = \frac{16 + 17 + 10 + 13 + 20 + 18 + 13 + 14 + 18}{9}$$

$$\bar{x} = \frac{139}{9}$$

$\bar{x} = 15.444$ (rounded to three decimal places)

Descriptive Statistics

- **Measures of Central Tendency**
- **Median:** value separating the higher half from the lower half of a data sample, a population, or a probability distribution. For a data set, it may be thought of as "the middle" value.
 - it is not skewed by a small proportion of extremely large or small values



Descriptive Statistics

- **Measures of Central Tendency**
- **Mode:** the value that appears most often in a set of data value.

How to Calculate the Mode:

Example Values

12, 7, 17, 9, 16, 29, 8, 4, 15, 7, 29, 6, 17, 2, 5, 12, 8, 17, 6, 6, 17, 17

Step 1:

Sort all the values from low to high.

2, 4, 5, 6, 6, 6, 7, 7, 8, 9, 12, 12, 15, 16, 17, 17, 17, 17, 17, 29

Step 3:

Find the value that occurs the most times. This is the **mode**.

2,	4,	5,	6,	6,	6,	7,	7,	8,	9,	12,	12,	15,	16,	17,	17,	17,	17,	17,	29
1	1	1	3			2		1	1	2		1	1			5			1

Mode = **17**



Descriptive Statistics

- Measures of Central Tendency

Mean, Median, Mode and Range

www.cazoommaths.com

Mean

Add all the numbers then divide by the amount of numbers

9, 3, 1, 8, 3, 6

$$9 + 3 + 1 + 8 + 3 + 6 = 30$$

$$30 \div 6 = 5$$

The mean is 5

Median

Order the set of numbers, the median is the middle number

9, 3, 1, 8, 3, 6

1, 3, 3, 6, 8, 9

The median is 4.5

Mode

The most common number

9, 3, 1, 8, 3, 6

The mode is 3

Descriptive Statistics

- **Measures of Dispersion**
- **Standard Deviation:** measure of the amount of variation or dispersion of a set of values.
- A low standard deviation indicates that the values tend to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation

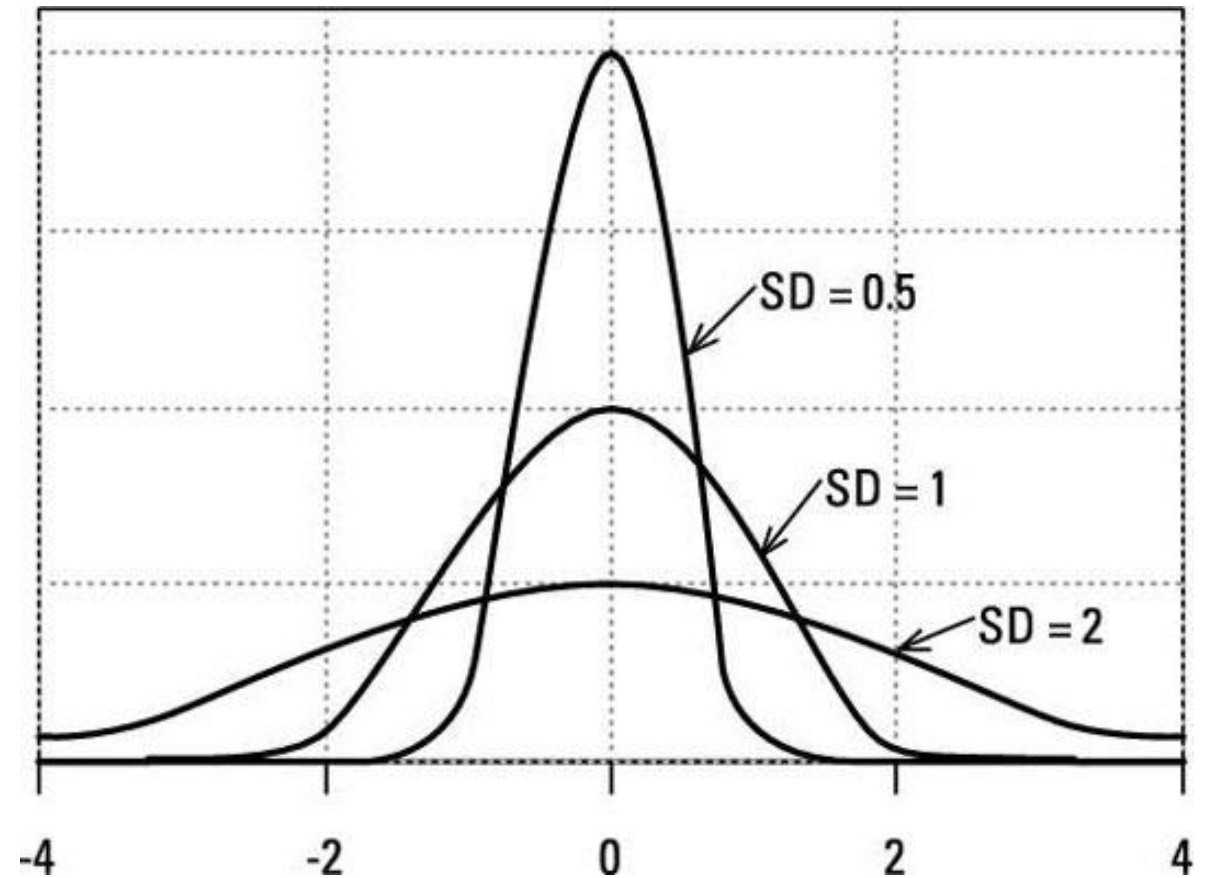
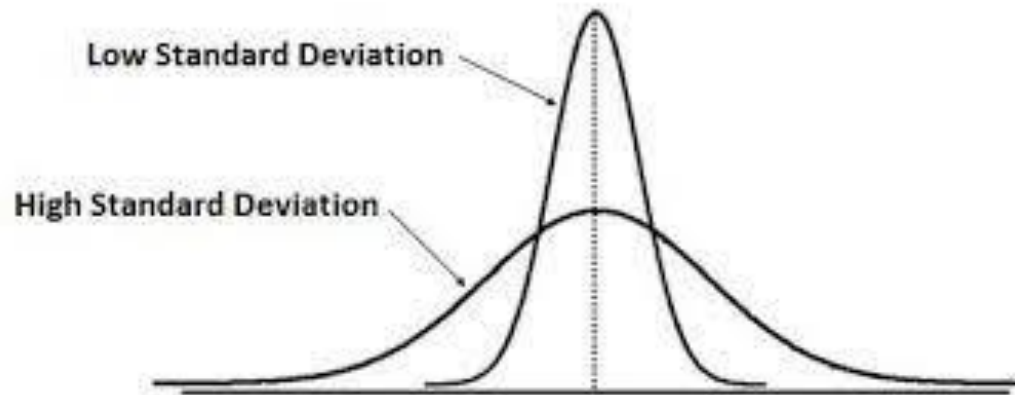
N = the size of the population

x_i = each value from the population

μ = the population mean

Descriptive Statistics

- Measures of Dispersion
- Standard Deviation:

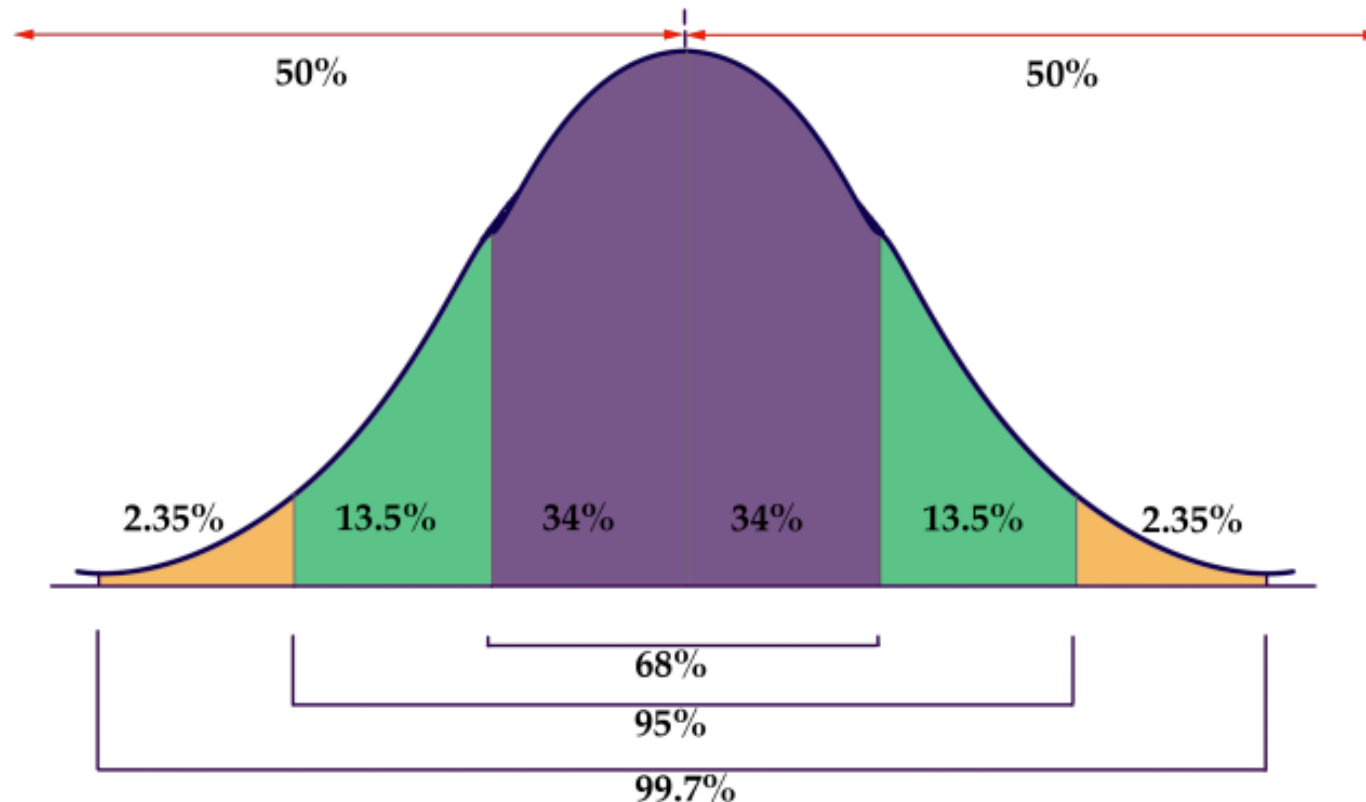


Descriptive Statistics

- Measures of Dispersion
- Standard Deviation:
 - Rules for normally distributed data:
 - If a data distribution is approximately normal then:
 - **68%** are within one standard deviation of the mean (mathematically, $\mu \pm \sigma$, where μ is the arithmetic mean);
 - **95%** are within two standard deviations ($\mu \pm 2\sigma$);
 - **99.7%** lie within three standard deviations ($\mu \pm 3\sigma$).
 - This is known as the **68-95-99.7 rule**, or the empirical rule.

Descriptive Statistics

- Measures of Dispersion
- Standard Deviation:
 - Rules for normally distributed data:



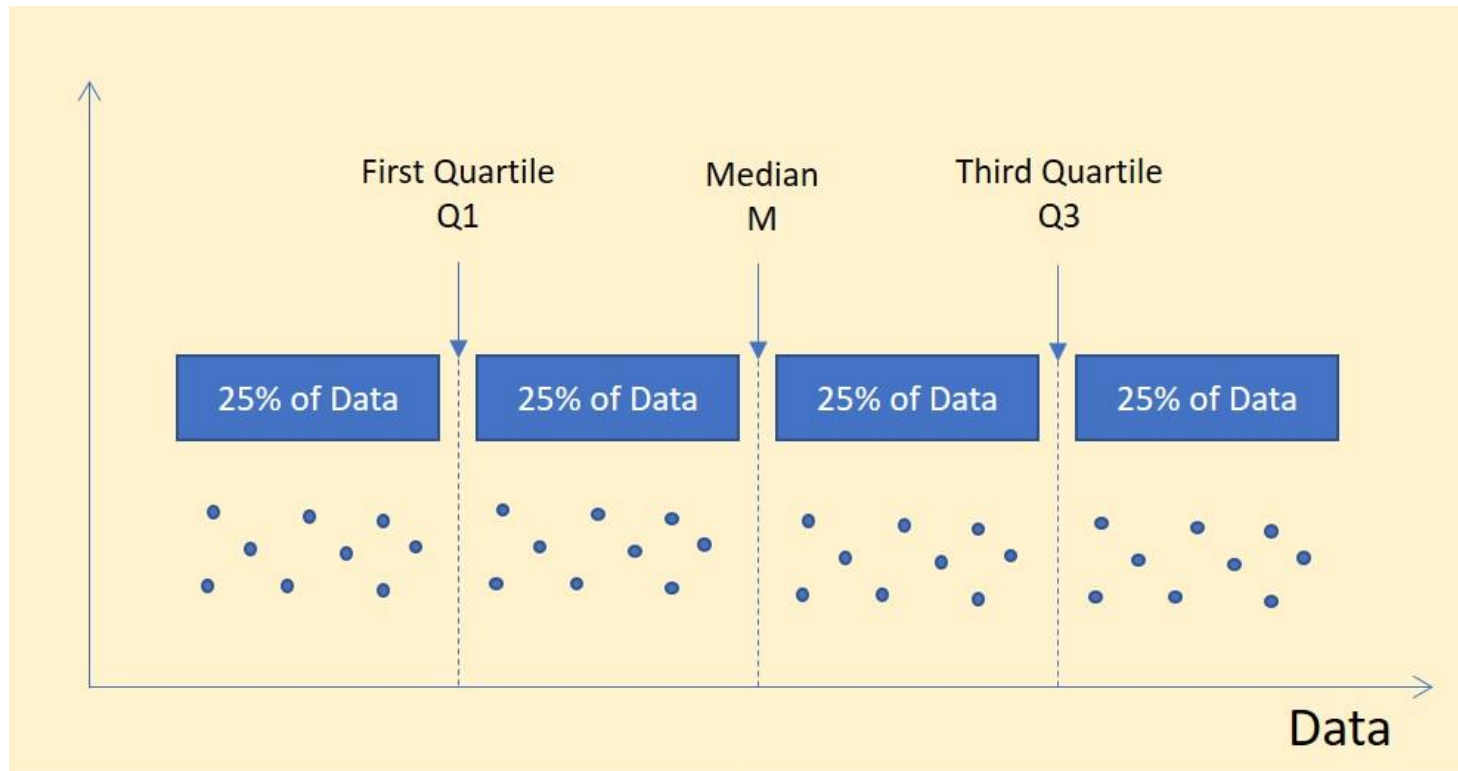
<https://courses.lumenlearning.com/math4libarts/chapter/the-empirical-rule/>

Descriptive Statistics

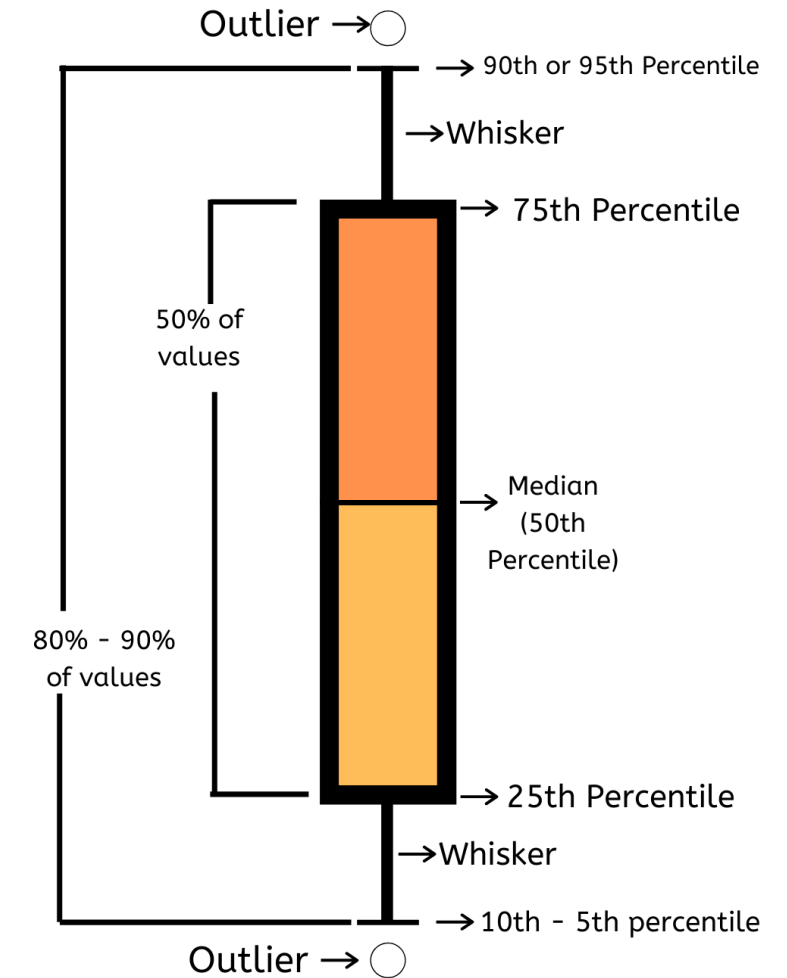
- **Measures of Dispersion**
- **Quartile:** divides the number of data points into four parts, or quarters, of more-or-less equal size. The data must be ordered from smallest to largest to compute quartiles; as such, quartiles are a form of order statistic. The three main quartiles are as follows:
 - **first quartile (Q1):** the middle number between the smallest number (minimum) and the median of the data set. It is also known as the lower or 25th empirical quartile, as 25% of the data is below this point.
 - **second quartile (Q2):** the median of a data set; thus 50% of the data lies below this point.
 - **third quartile (Q3):** the middle value between the median and the highest value (maximum) of the data set. It is known as the upper or 75th empirical quartile, as 75% of the data lies below this point

Descriptive Statistics

- Measures of Dispersion
- Quartile:



<https://www.sigmamagic.com/blogs/what-are-quartiles/>

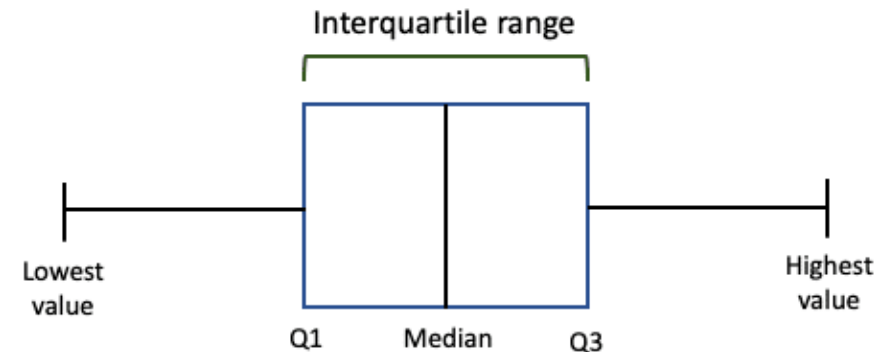
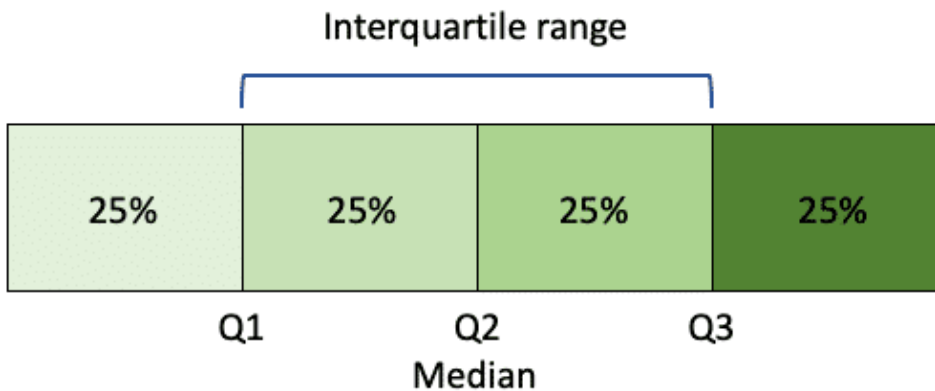


By: Vinita Silaparasetty

<https://medium.com/@vinitasilaparasetty/quartiles-for-beginners-in-data-science-2ca5a640b07b>

Descriptive Statistics

- **Measures of Dispersion**
- **Interquartile Range (IQR):** a measure of statistical dispersion, which is the spread of the data. The IQR may also be called the midspread, middle 50%, or H-spread. It is defined as the difference between the 75th and 25th percentiles of the data.



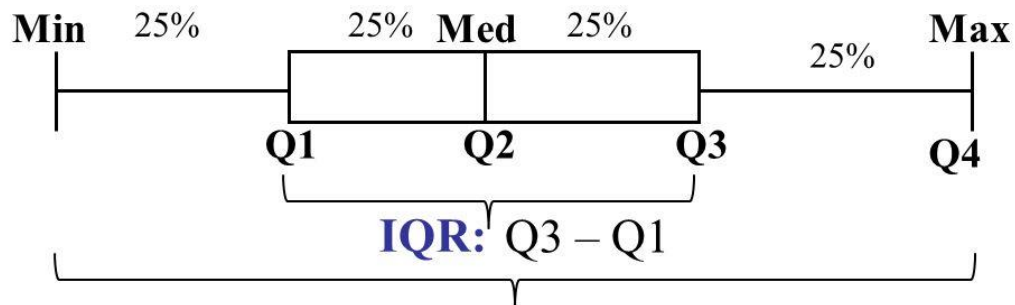
Descriptive Statistics

• Measures of Dispersion

Boxplot: “Box and Whisker”

Whiskers represent outside quartiles (Min to Q1 and Q3 to Max)

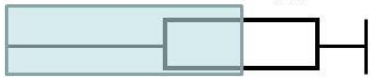
Boxes Represent inside quartiles (Q2 to Med and Med to Q3)



Range: Max – Min

Skewed Left (negatively):

Less data to the left (spread out)

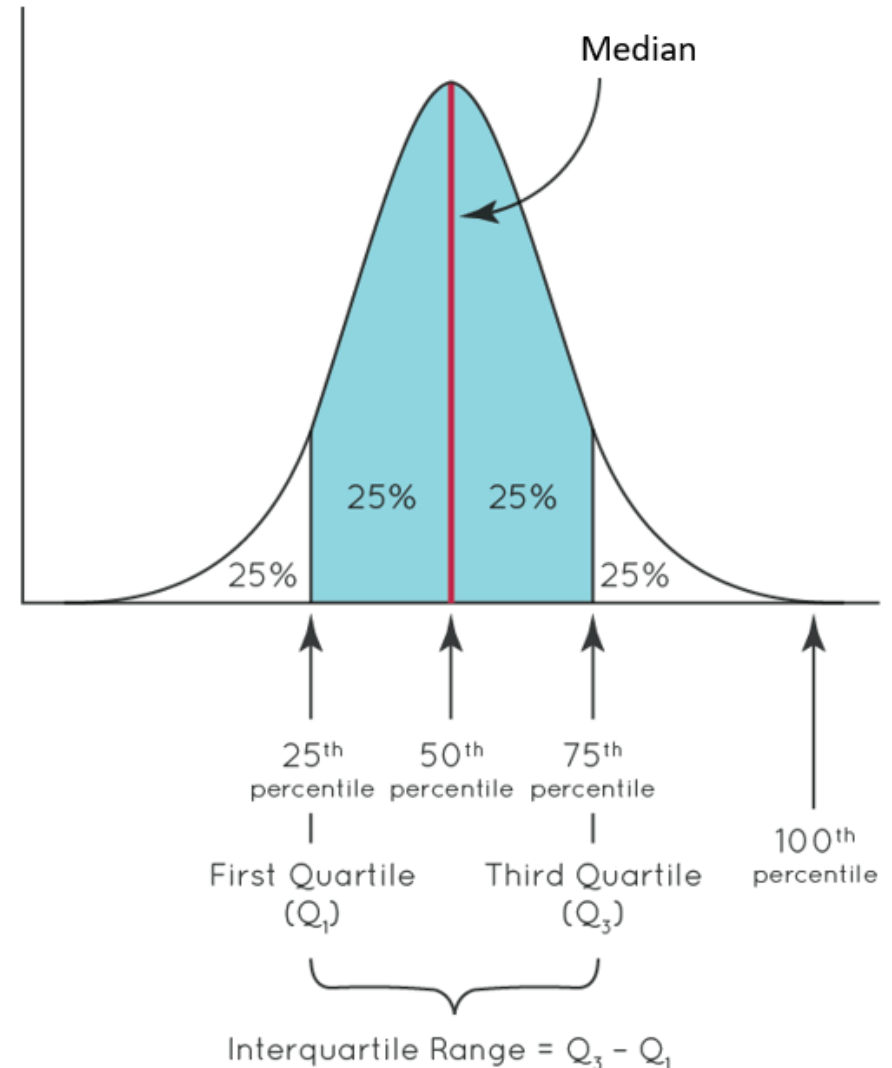


Skewed Right (positively):

Less data to the right (spread out).



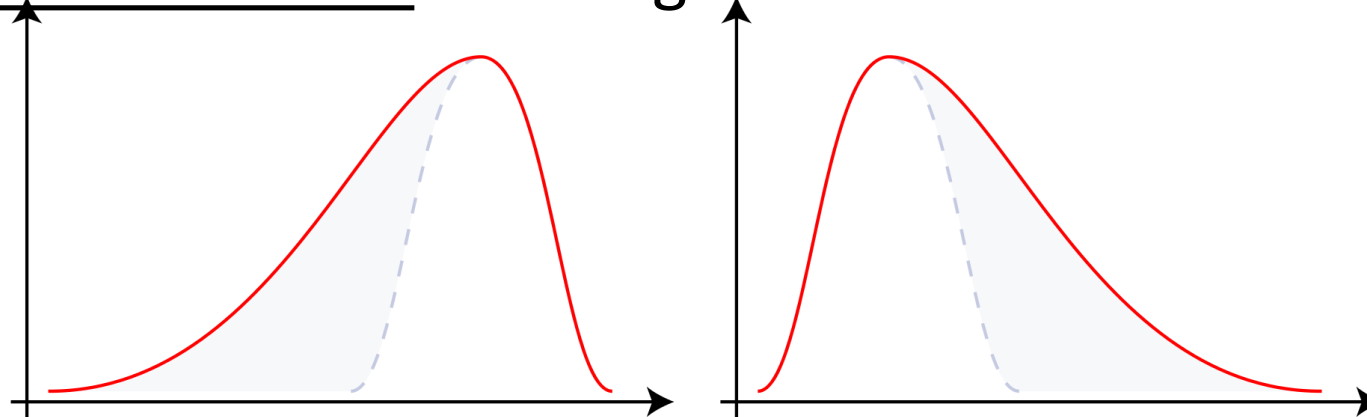
<https://slideplayer.com/slide/7864524/>



<https://www.cuemath.com/quartile-formula/>

Descriptive Statistics

- **Measures of Distribution**
- **Skewness:** measure of the asymmetry of the probability distribution of a real-valued random variable about its mean.
 - **negative skew:** the left tail is longer; the mass of the distribution is concentrated on the right of the figure.
 - **positive skew:** The right tail is longer; the mass of the distribution is concentrated on the left of the figure.



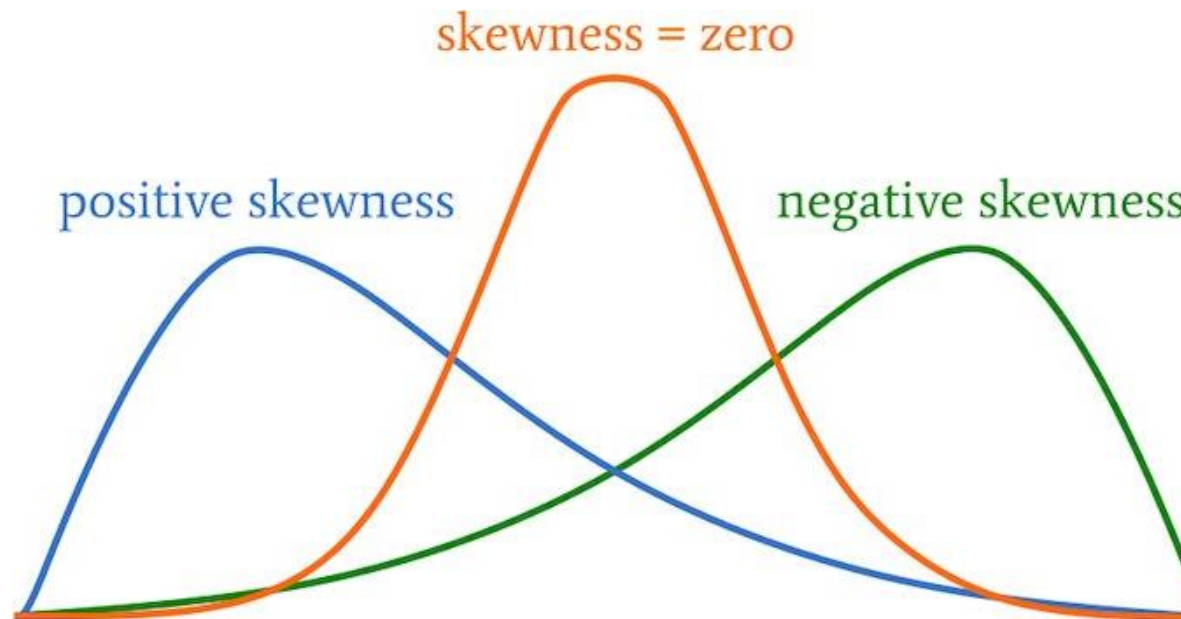
Negative Skew

Positive Skew

Descriptive Statistics

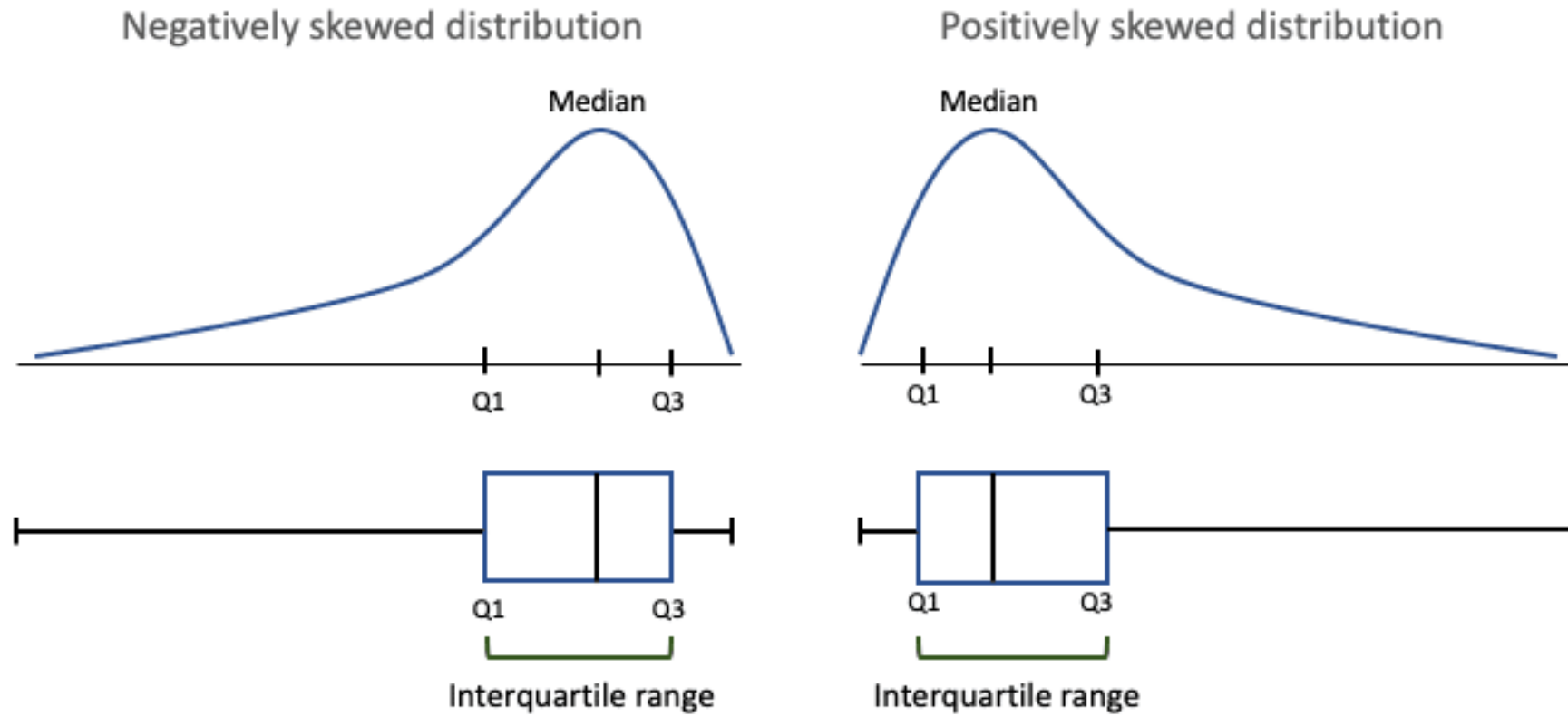
- Measures of Distribution
- Skewness:

Uma variável é simétrica se a divisão do coeficiente de skewness pelo seu erro padrão estiver situado entre **-1,96** e **1,96**



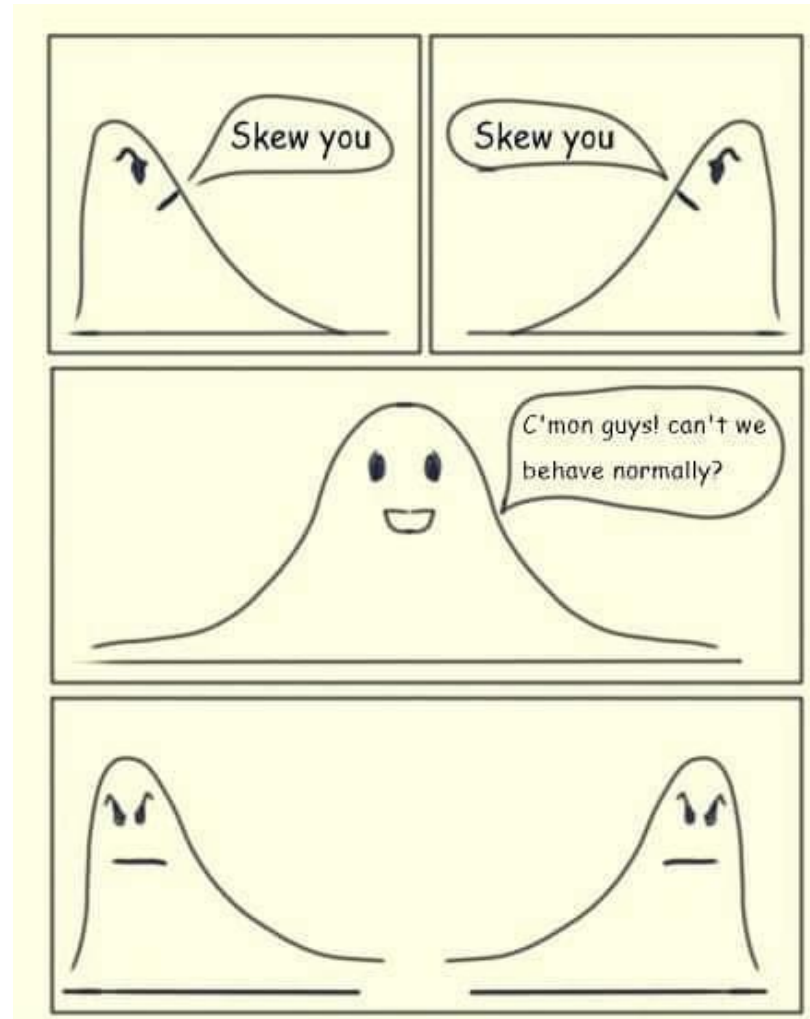
Descriptive Statistics

- Measures of Distribution
- Skewness:



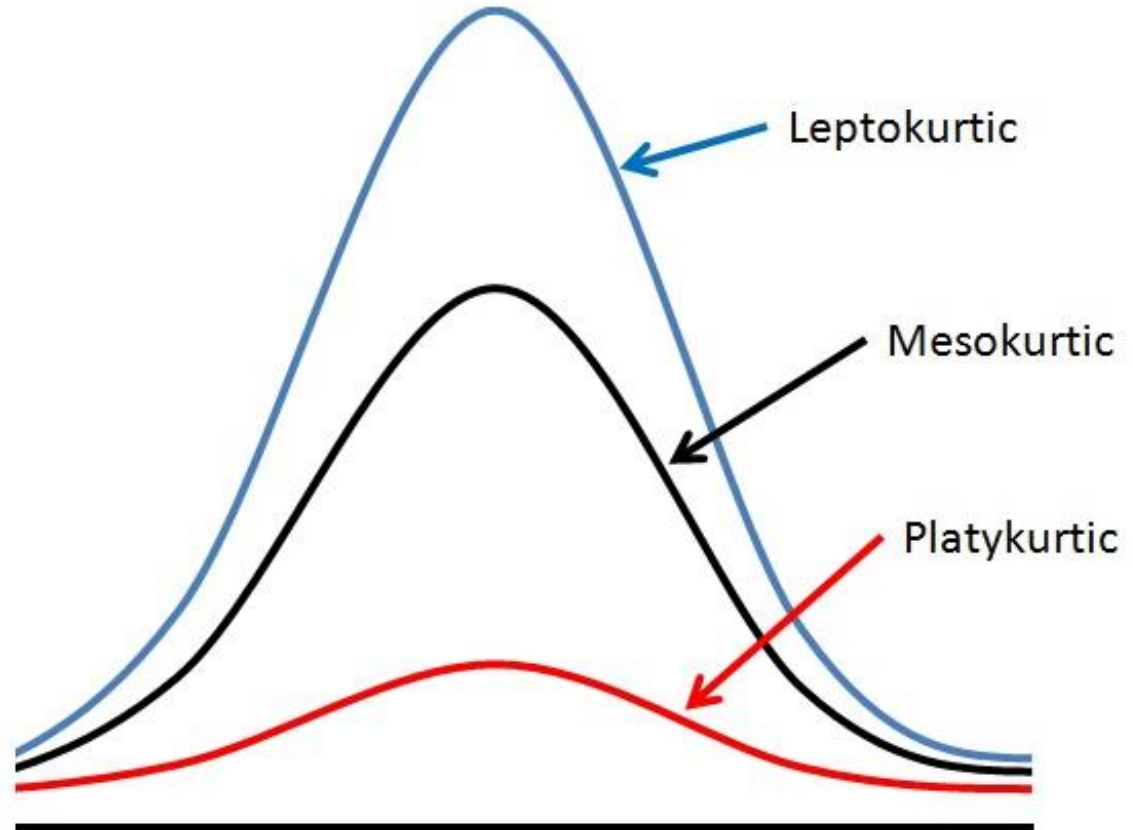
Descriptive Statistics

- Measures of Distribution
- Skewness:



Descriptive Statistics

- **Measures of Distribution**
- **Kurtosis:** measure of relative peakedness of a distribution (literally the amount of hump). It is a shape parameter that characterizes the degree of peakedness.
- The kurtosis of any univariate normal distribution is **3**.

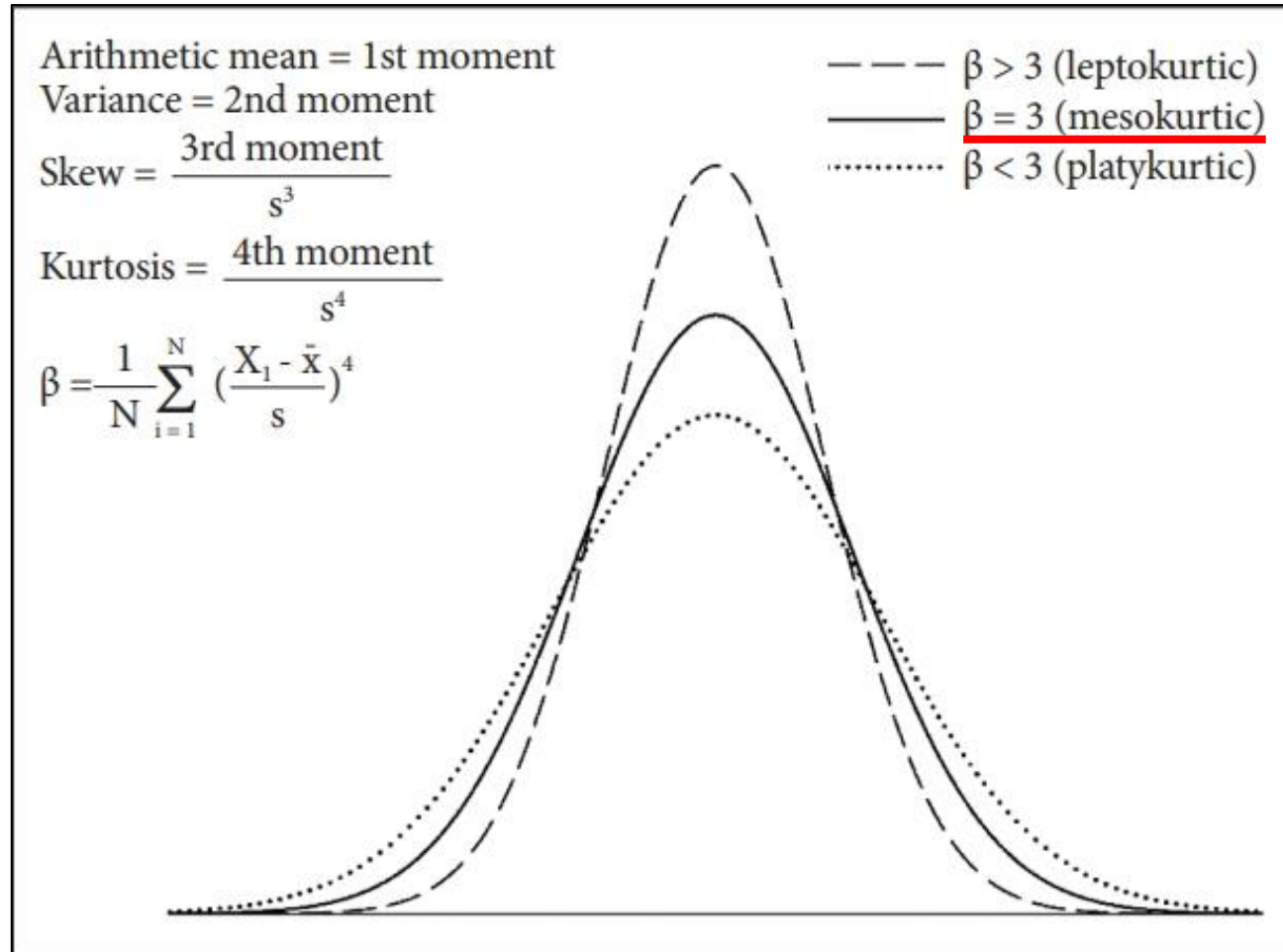


Descriptive Statistics

Uma variável é mesocúrtica se a divisão do coeficiente kurtosis for aproximadamente 3

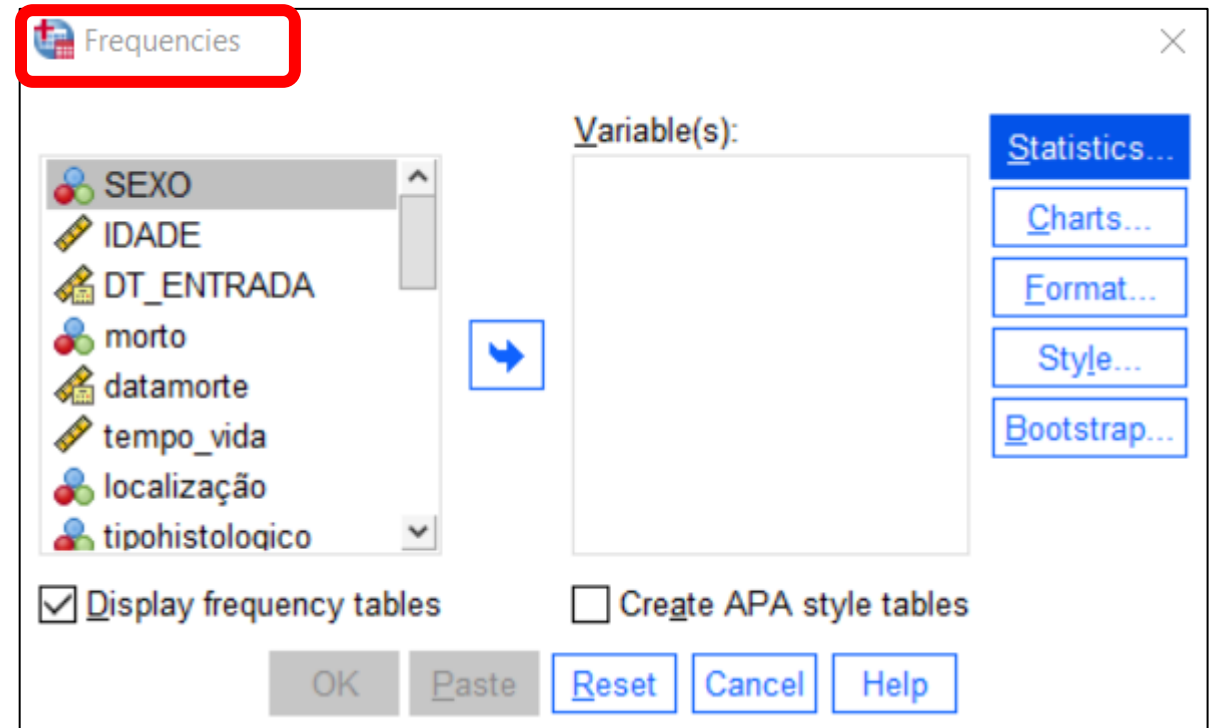
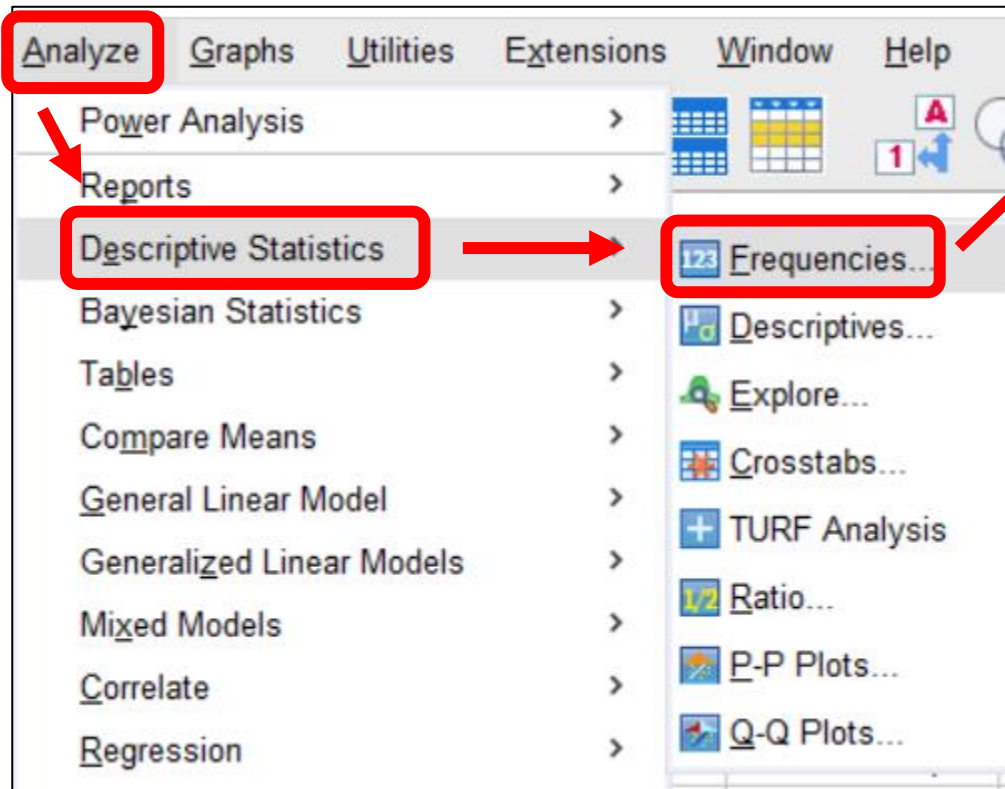
- Measures of Distribution

- Kurtosis:



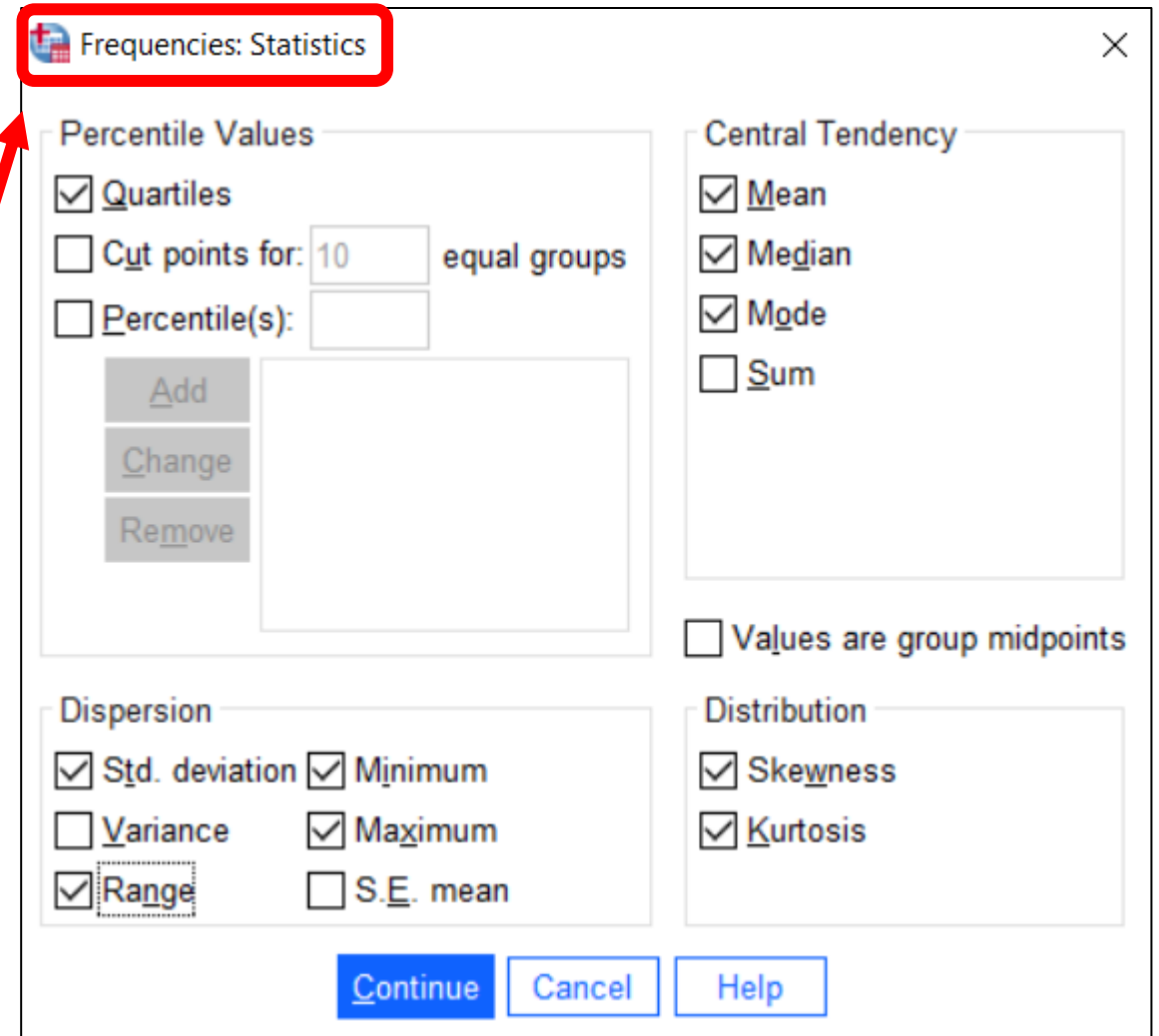
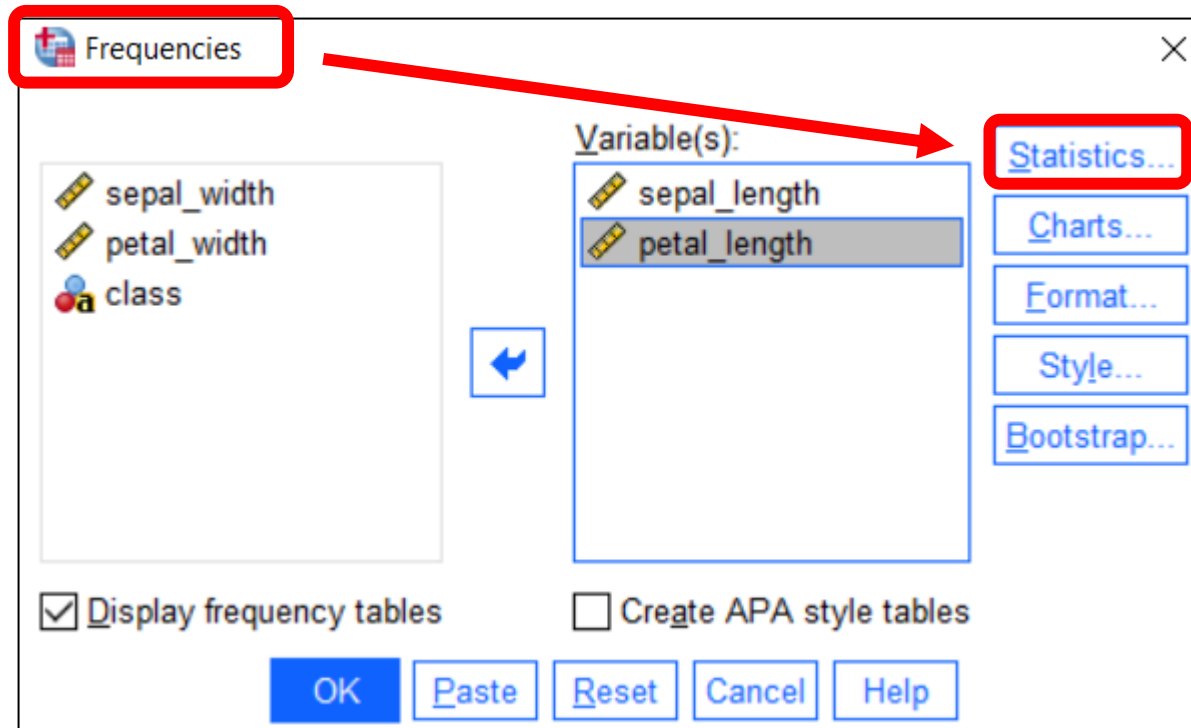
Descriptive Statistics

- **SPSS**



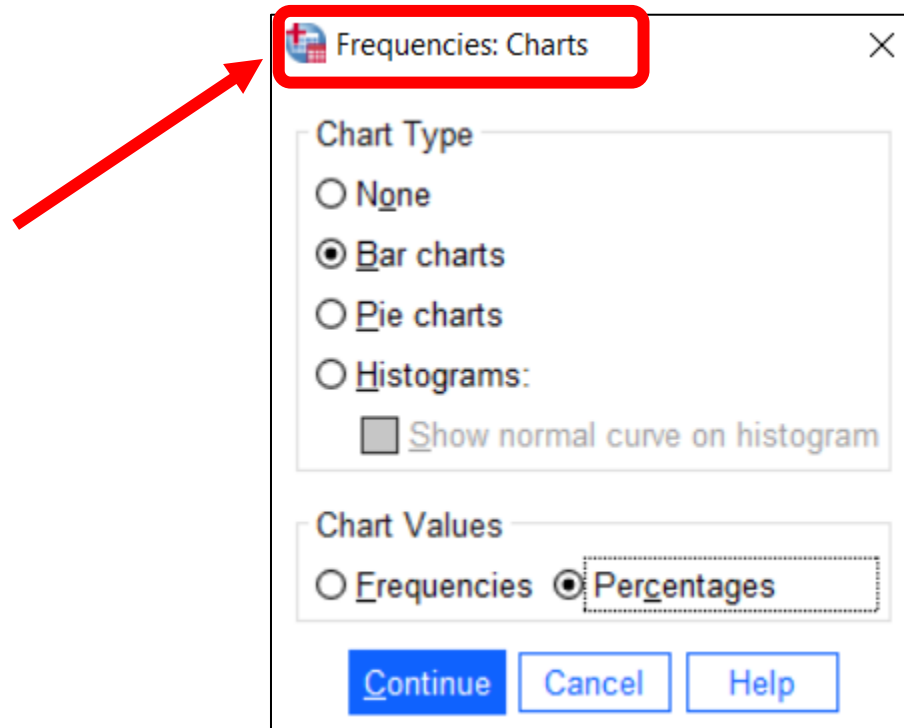
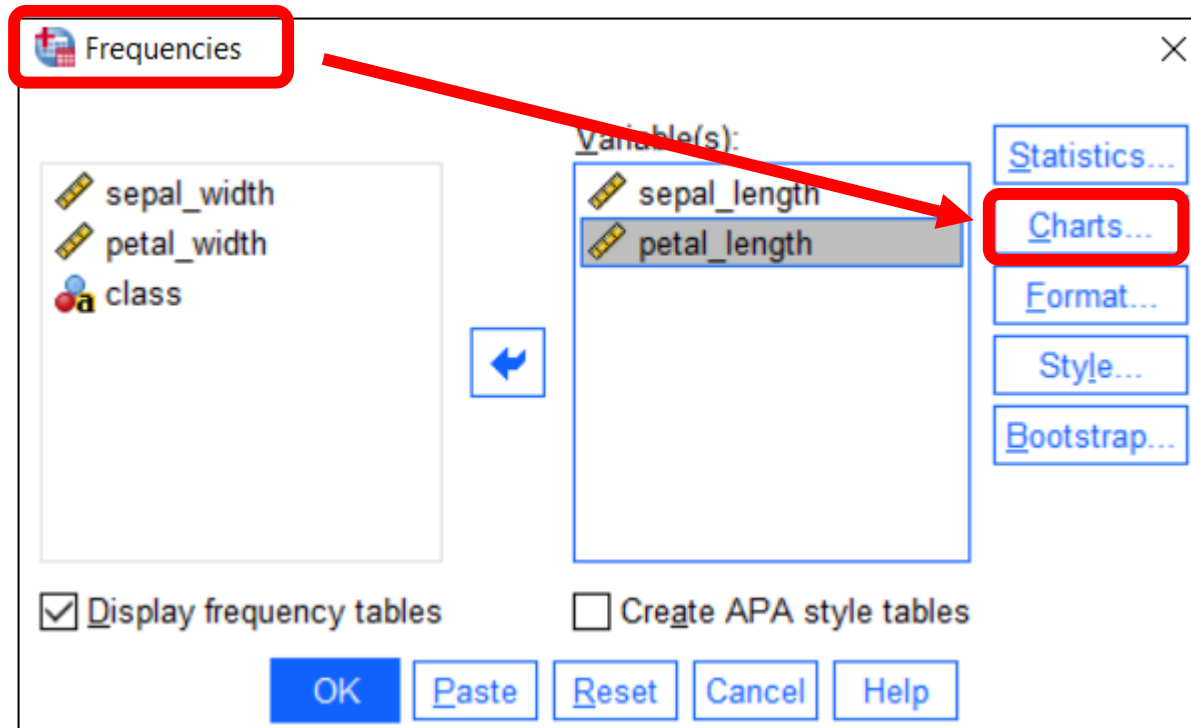
Analyse » Descriptive Statistics » Frequencies

Descriptive Statistics



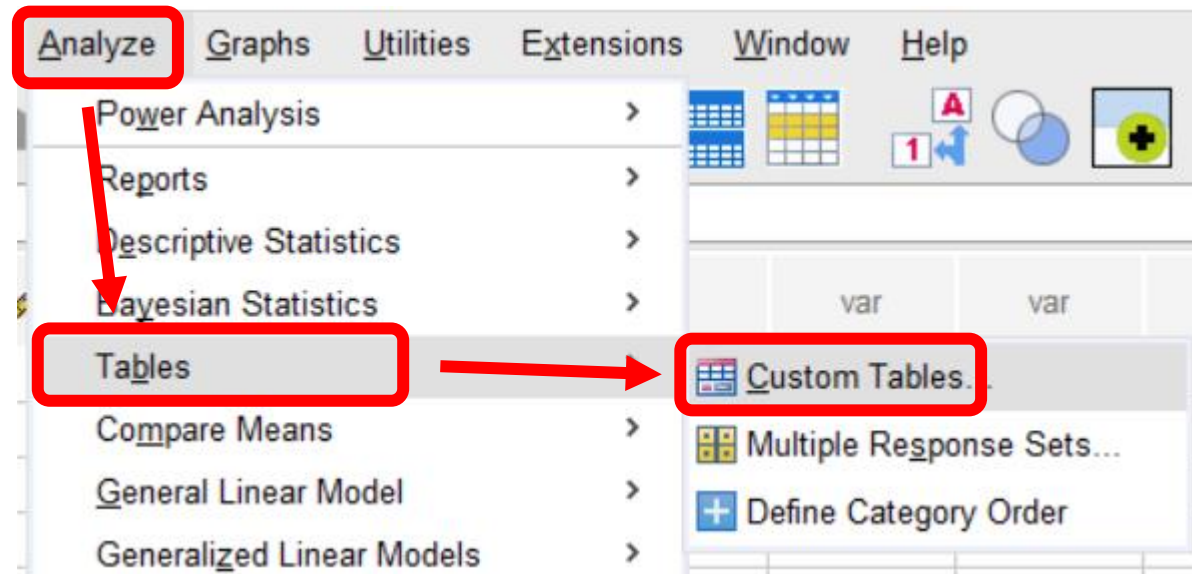
- **SPSS**

Descriptive Statistics



Descriptive Statistics

- **SPSS - Custom Tables**
- enables to summarize statistics data, and distinguishes between scale variables and categorical variables



Analyse » Tables » Custom Tables

Descriptive Statistics

- **SPSS - Custom Tables**

Categorical
variable

The image shows the 'Custom Tables' dialog box in SPSS. The 'Variables' list on the left contains 'sepal_length'. The 'Categories' list on the left contains 'Category 1' and 'Category 2'. The 'Columns' list on the right contains 'petal_width'. The 'Rows' list on the left contains 'class'. The 'Summary Statistics' section at the bottom right shows 'Position' set to 'Columns' and 'Source' set to 'Column Variables'. The 'Define' section at the bottom left shows 'N% Summary Statistics...' selected. The 'Category Position' dropdown is set to 'Default'. The 'OK' button is highlighted.

	Mean	Median	Mode	Range	Std. Deviat...
Category 1	nnnn,n	nnnn,n	nnnn,n	nnnn,n	nnnn,n
Category 2	nnnn,n	nnnn,n	nnnn,n	nnnn,n	nnnn,n

Scale
variable

Anaconda Distribution

- **Bibliography**

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